

PATIENT AND PHYSICIAN EVALUATION OF DIGITAL HEALTH DECISION SUPPORT TOOL IN COPD CARE

Jeenat Mehareen¹, Megan Simmons², Phalgun Joshi³, Mohsen Sadatsafavi¹

1 Respiratory Evaluation Sciences Program, Faculty of Pharmaceutical Sciences, University of British Columbia, Vancouver, Canada
2 Faculty of Medicine, University of British Columbia, Vancouver, Canada
3 The Legacy for Airway Health and Centre for Lung Health, Vancouver Coastal Health Research Institute, Vancouver, Canada

UBC Faculty of Pharmaceutical Sciences

WHY

Risk prediction only changes care if the people using it; **clinician and patient**; find it usable.

The IMPACT stepped-wedge cluster RCT evaluated a personalized COPD exacerbation prediction tool (ACCEPT) at two Vancouver teaching clinics. It also provided a tailored information sheet to support treatment discussions.

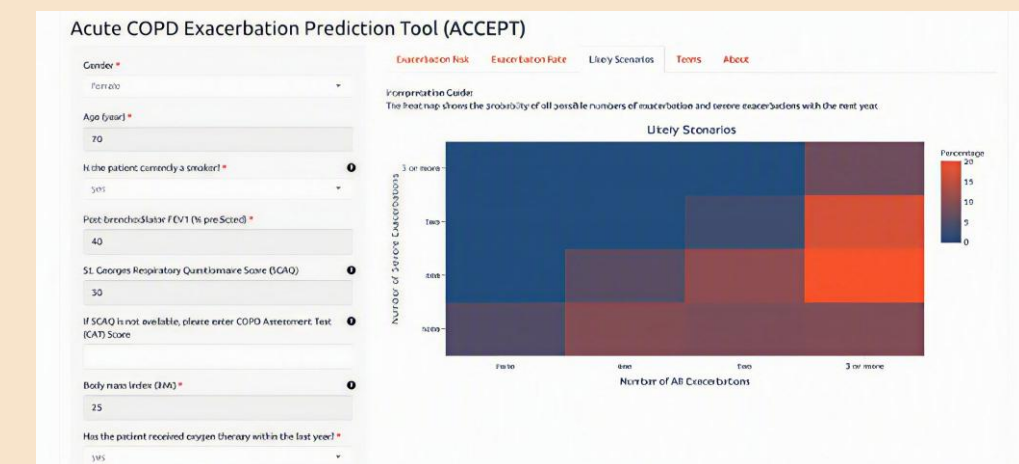
We evaluated patients' perceptions of this sheet and physicians' usability assessments of the tool.

HOW

- **ADI** (ACCEPT Decision Intervention): ACCEPT 2.0 exacerbation risk prediction plus Canadian Thoracic Society pharmacotherapy algorithms, inside the electronic health record.
- **Information Sheet**: Individualized to the patient's lung function, predicted risk and prescribed medication; six sections.

6 physicians

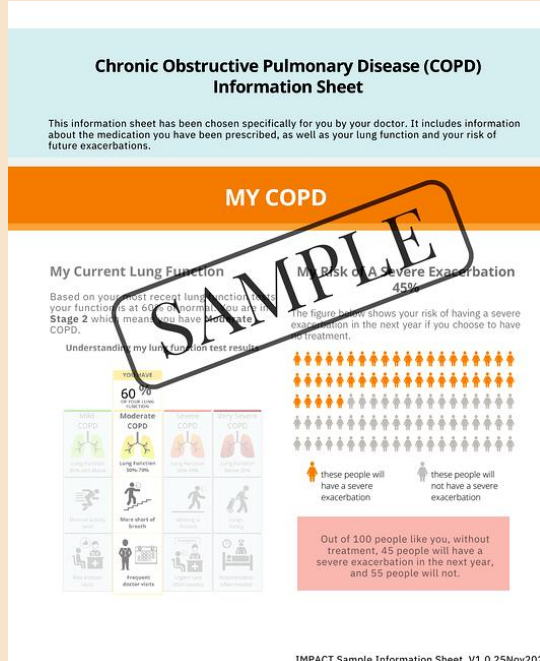
completed the System Usability Scale and the Technology Acceptance Model (PU, PEOU; 1–7), with free-text questions



ADI: FOR THE PHYSICIAN

72 patients

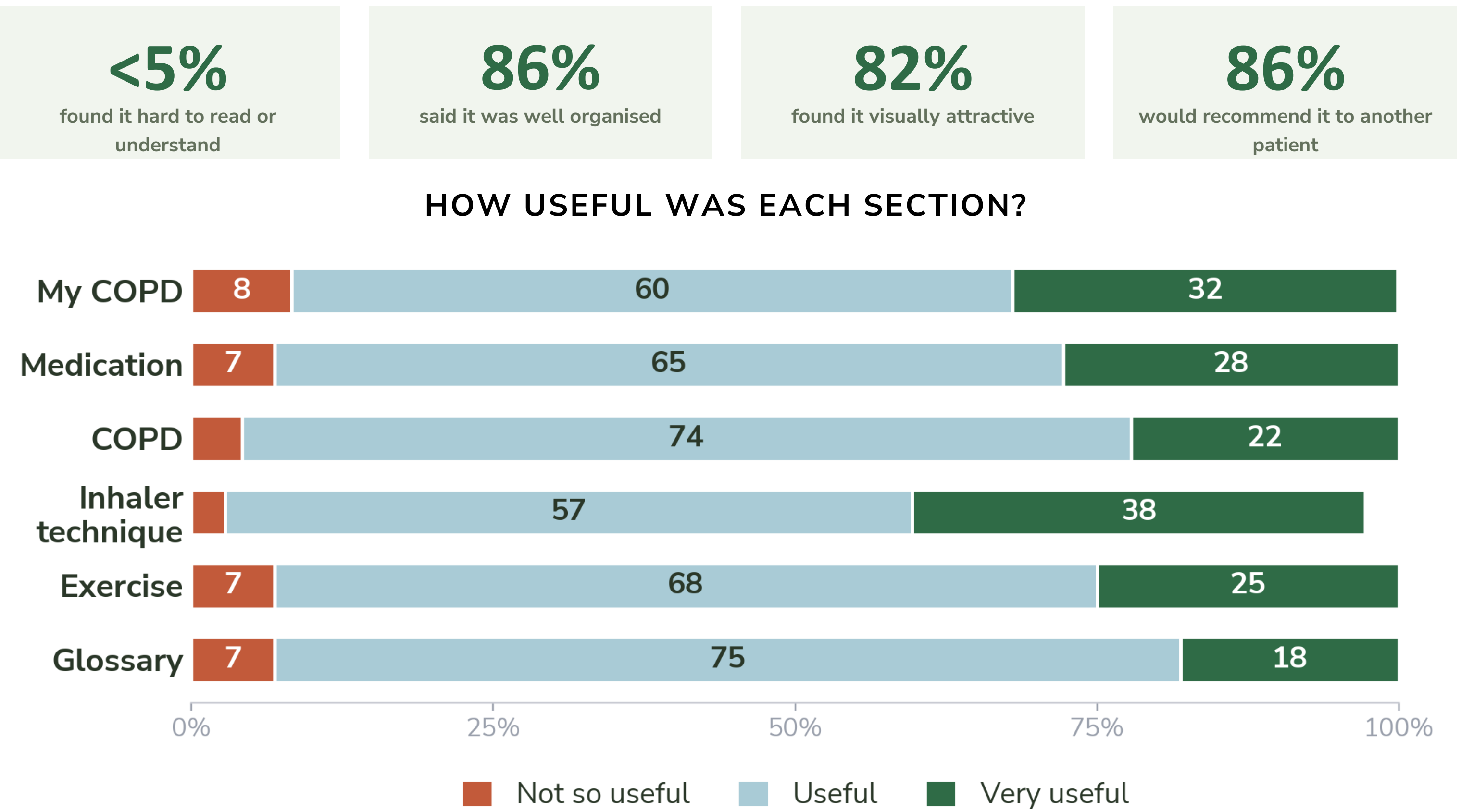
rated usefulness and amount of information in each section, plus readability, design and organization.



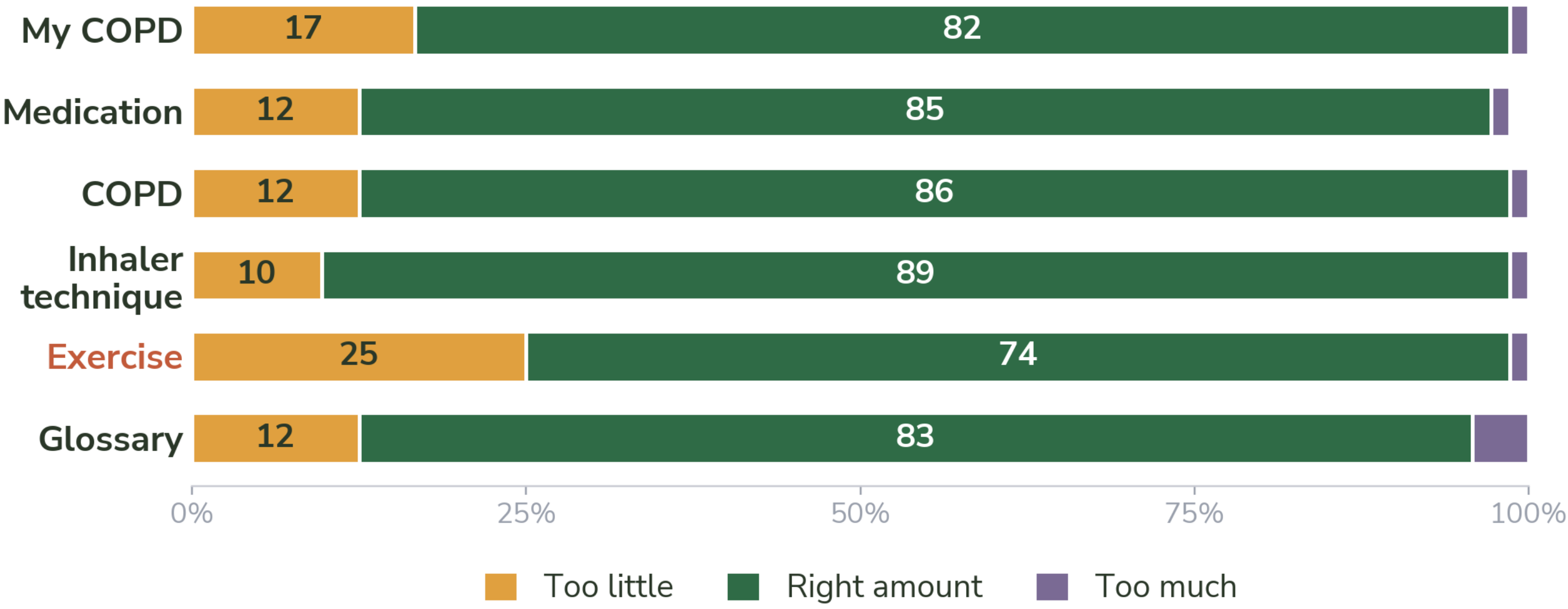
INFORMATION SHEET: FOR THE PATIENT

Both surveys were completed anonymously at the end of the study.

WHAT PATIENTS SAID



WAS THERE THE RIGHT AMOUNT OF INFORMATION?

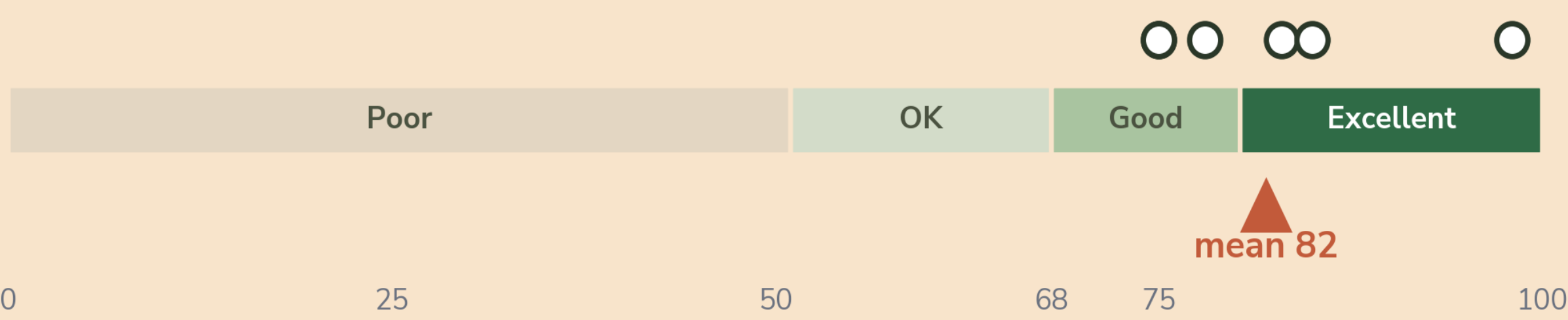


Every section was rated useful or very useful by **92–96%** of patients.

WHAT PHYSICIANS SAID

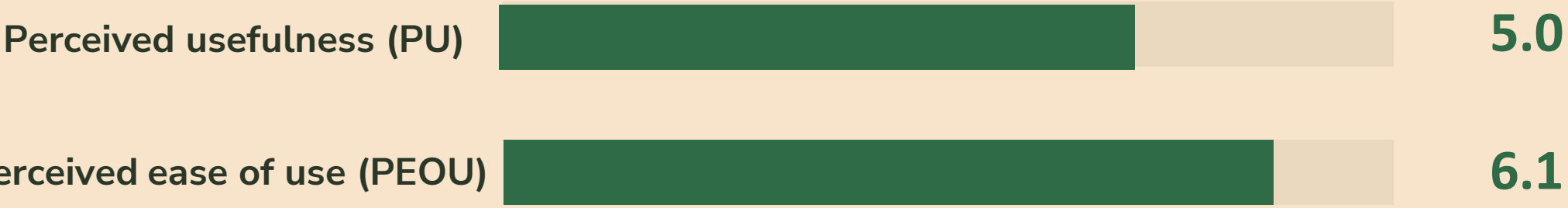
SYSTEM USABILITY SCALE (N=6)

Each circle presents one physician



A mean SUS of **82** is a grade A; which is well above the 68-point benchmark for an average system, with every physician at or above 75.

TECHNOLOGY ACCEPTANCE MODEL (1–7)



IN THEIR WORDS

- “ Easy to use, quick, validated and an effective complement to clinical assessment.
- “ I like the integration with guideline recommendations that helps me in guiding best practices for my patients.
- “ It was easy to use and the generation of recommended therapies based on CTS guidelines was useful.
- “ I like the easy radio buttons and standardized text output to paste into my note.

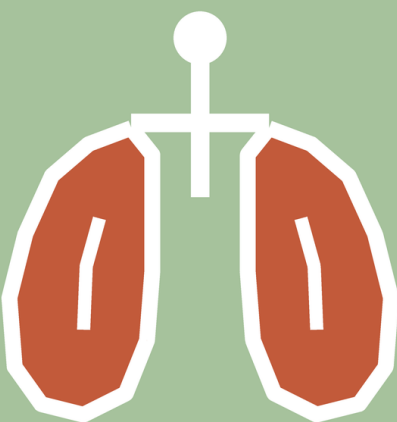
Physicians wanted **biologics information** added in ACCEPT and the tool **extended to other chronic respiratory conditions**.

DID THE INFORMATION SHEET HELP?



SUS 82 = GRADE A USABILITY

TAKE AWAY POINTS



1. Both patient and physician accepted the intervention: patients found the sheet readable, well designed and worth recommending (**86%**), and physicians rated the ADI grade-A usable (SUS **82**) and easy to use (PEOU **6.1/7**).
2. Exercise guidance is the clearest place to build on next; a quarter of patients wanted more of it, and it was also where the sheet moved behaviour most (**42%** accessed exercise resources versus **11%** for smoking cessation).
3. These findings show the feasibility and acceptability of integrating precision medicine tools with patient-centered communication into routine COPD care.



ERS
E-POSTER